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The Distributive Property

The Distributive Property

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The Distributive Property

The **distributive property** involves multiplication and addition. Study this example to understand how the property works.

Example 1

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At a dollar store, all items are \$1.10. Maia buys 7 plastic tablecloths and 3 packs of paper plates. How much money did she spend?

Method 1

Total items: $7 + 3 = 10$

Multiply by cost: $10 \times \$1.10 = \11.00

Method 2

Cost of tablecloths: $7 \times \$1.10 = \7.70

Cost of plates: $3 \times \$1.10 = \3.30

Total cost: $\$7.70 + \$3.30 = \$11.00$

In the example, both methods gave a final answer of \$11. How would you describe the difference between the methods? Method 1 adds first. Method 2 multiplies first. The distributive property describes this relationship between adding and multiplying.

Instead of adding b and c and then multiplying by a , you can distribute a to both b and c and then add.

$a(b + c) = ab + ac$

Note: Writing two variables next to each other indicates multiplication: $a(b + c)$ means $a \times (b + c)$, and $ab + ac$ means $(a \times b) + (a \times c)$.

You can also use the distributive property with subtraction.

$a(b - c) = ab - ac$

Example 2

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Carol used to earn \$11 per hour. Now she earns \$16 per hour. How much more does she earn now for a 40-hour work week?

Method 1

$40 \times (\$16 - \$11)$

$= 40 \times \$5$

$= \$200$

Method 2

$(40 \times \$16) - (40 \times \$11)$

$= \$640 - \440

$= \$200$

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Lesson 2: Expressions

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